2016 EXAMINATIONS

Lancaster University

Part II

COMPUTING AND COMMUNICATIONS [2 Hours]

SCC.212 Advanced Programming

Candidates are asked to answer THREE questions from FOUR; each question is worth a total of 20 marks.

Use a separate answer book for each question.

Question 1

1.a Object oriented languages such as Java employ a number of mechanisms that allow the construction of elegant programs. For each of the following mechanisms, describe the OO principle that it exhibits, and outline its main role within the language.

- the extends keyword
- the implements keyword
- the class called Class **[6 marks]**

1.b Imagine you have been asked to write a Java program to model the popular children's card game **Top Trumps**. This simple game involves two players and a deck of cards that are themed around a particular topic. Each card contains quantified information about an object related to that theme. All cards in a deck contain the same type of information. For example, Top Trumps Cars might contain a deck of cards each with the following categories of information:

- The name of a particular car
- Its top speed
- Its acceleration
- Its cost

Players have a collection of cards, and on each turn take their first card and one player chooses a category. The players then compare the values for the chosen category, and the player with the highest value in that category is declared the winner.

i. Given that every category has a name, and either an integer, or a floating point value, show how you could create three Java classes in an inheritance relationship to elegantly represent a category. Write Java code to define each of your classes, including all necessary instance variables and constructors.

[5 Marks]

ii. Show how you could write a method capable of comparing two categories to determine if an instance of a given category contains a higher value than another. You should not permit integer categories to be compared with floating point categories.

[3 marks]

iii. Write a class to represent a Card that contains a collection of categories. There may be any number of categories on a card, and those categories need not all be of the same type. You should include a method that allows a given category to be added to a card.

[3 marks]

iv. Write a method for your Card class that, when given another card and category name, determines which card is the winner.

[3 marks]

Total 20 Marks

Question 2

2.a Immutability is arguably the most significant difference between procedural and functional programming languages. Explain what is meant by the term immutability, its benefits and drawbacks, and compare and contrast the differing approaches to immutability found in Erlang and Java.

[6 marks]

2.b Imagine a simple game where a player has two six sided dice. The two dice are rolled and the face values are simply added together to determine a score.

A game is played, and the following rolls are recorded:

Dice1	3	1	5	2	3	6	4	5	5	3
Dice2	3	6	1	2	5	6	3	2	6	1

i. Write Erlang statements to create one or more lists to represent the sequence of dice throws shown above.

[2 marks]

ii. Write an Erlang function that, from first principles (without the use of built in functions), takes the list(s) from part i) and returns a list that contains the totals of each round. E.g. in the example above this would be 6, 7, 6, 4, 8 etc.

[4 marks]

iii. Write a further Erlang function that, given the result from ii), calculates the sum of all the values in that list.

[4 marks]

iv. Describe what is meant by the term "tail recursive function", and state whether or not your solution to iii) is tail recursive. If it is not, write a tail recursive equivalent.

[4 marks]

Total 20 Marks

Question 3

3.a Briefly explain the operation of the **filter()** and **reduce()** functions in Python.

[**2 marks**]

3.b What is the final result of running the following Python programs?

i.

```
def f(n, m):
    return lambda x,y: n*x +m*y
g =  f(2, 4)
print(g(3,2))
```

ii.

```
s = 'Lancaster'
p = range(1,len(s)-1, 2)
p = list(p)
print(p)
```

iii.

```
s = 'See you in the morning'
s = s[-22:-18]+s[8:]
print(s)
```

[**6 marks**]

3.c Consider the following list of student marks:

marks = [23, 34, 48, 48, 52, 55, 56, 59, 60, 64, 65, 67, 69, 72, 78, 90]

The table below shows how grades are calculated from a range of marks:

Marks range	Grade
70 - 100	A
60 - 69	B
50 - 59	C
40 - 49	D
0 - 39	F

i. Write in Python a function called *grade* that accepts a mark and returns the mark and its corresponding grade. For example, a mark of 60 would return '60 B'.

ii. Write a Python *map* function that generates a list of marks together with their corresponding grades.

iii. Write a Python *reduce* function that computes the mean mark from the list of marks.

iv. Write a *filter* function that uses a lambda to return a list of marks that are within -10% and +10% of the mean mark.

[**12 marks**]

Total 20 Marks

Question 4

4.a Polymorphism refers to a programming language's ability to redefine methods for derived classes. A common way to implement polymorphism is through method overriding as shown in the Java program below. Rewrite this *Java* program in *Python* to achieve the same result. Provide brief comments on the different parts of your program.

```
public abstract class Shape
{
    double a, b;
    Shape(double d1, double d2) {
    a = d1;
    b = d2;
    }
    abstract double area();
}

public class Circle extends Shape
{
    Circle(double radius) {
      super(radius, 0);
    }

    double area() {
      return Math.PI * a * a;
    }
}

public class Triangle extends Shape
{
    Triangle(double base, double height){
        super(base, height);
    }
    double area(){
        return 0.5*(a * b);
    }
}

class Rectangle extends Shape
{
    Rectangle(double length, double breadth)
    {
        super(length, breadth);
    }

    double area()
    {
        return a * b;
}
```

Question 4 continues on next page...

Question 4 continued...

```
public class AreaExample
{
    public static void main(String[] args)
    {
        Shape shape = new Circle(5);
        System.out.println("Area of circle =" +" " +
         shape.area());
        shape = new Triangle(5, 10);
        System.out.println("Area of triangle = "+
         shape.area());

    }
}
```

[14 marks]

4.b Rewrite the following *Java* method as a *Python* **lambda** function

```
public boolean methodA(int x, y)
{
   return (x%6!=1 & y>4);
}
```

[2 marks]

4.c Consider the Python program shown below

```
h =  lambda x:x%2 == 0 and x%4!=0
g =  lambda x:x//6==0 or h(x)
f =  lambda x: h(x) and g(x)

valueList =[]

for x in range(2,9,2):
     if(f(x)==True):
         valueList.append(x)
print(valueList)
```

i. Describe the values returned by **f** when the program is executed.

[2 marks]

ii. Describe the values returned by **g** when the program is executed.

[2 marks]

Total 20 Marks

---End of Paper---